Lecture 26 (Farkas' Lemma)

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Today we will talk about Farkas' Lemma and will prove the result using two methods.

Theorem (Farkas' Lemma)

Let $A \in \mathbb{R}^{m \times n}$ and $\vec{b} \in \mathbb{R}^m$. Then exactly one of the following holds:
(a) $\exists \vec{x} \geq 0$ such that $A\vec{x} = \vec{b}$
(b) $\exists \vec{p}$ such that $\vec{p}^T A \geq 0$ and $\vec{p} \cdot \vec{b} < 0$
The above is called **Farkas' Lemma**.

Proof 1

We will show two parts of the proof. The first part will be shared in both versions of the LP. Suppose both **(a)** and **(b)** both hold. Then we assume $\exists \vec{x} \geq 0$ such that

$$A\vec{x} = \vec{b}$$

and $\exists \vec{p}$ such that $\vec{p}^T A \geq 0$ and $\vec{p} \cdot \vec{b} < 0$. Then we see

$$\vec{p}^T A\vec{x} = \vec{p}^T \vec{b} = \vec{p} \cdot \vec{b}$$

but as

$$\left(\vec{p}^T A\right), \vec{x} \geq 0$$

we see

$$\vec{p}^T A\vec{x} = \left(\vec{p}^T A\right)\vec{x} \geq 0$$

but

$$\vec{p}^T A\vec{x} = \vec{p} \cdot \vec{b} < 0$$

Thus to avoid contradiction, we cannot have both **a)** and **b)** be true at the same time. This implies that if **a)** holds, then **b)** cannot hold. Now, we will show that if **a)** doesn't hold, then **b)** must hold. This is the part that will be shown uniquely here for **Proof 1**. Suppose $\nexists \vec{x} \geq 0$ such that $A\vec{x} = \vec{b}$. Then we can consider a primal LP:

$$\text{maximize } \vec{0} \cdot \vec{x}$$

such that

$$A\vec{x} = \vec{b}$$

$$\vec{x} \geq 0$$

Then this primal LP yields the dual LP

$$\text{minimize } \vec{p} \cdot \vec{b}$$

such that

$$\vec{p}^T A \geq 0$$

$$\vec{p} \in \mathbb{R}^m$$

Since these LPs are dual and the primal LP is infeasible, this implies that the dual LP is either unbounded or infeasible. As $\vec{p} = 0$ is a feasible solution, the dual LP cannot be infeasible. As the dual is unbounded, then $\exists \vec{p}$ that is feasible such that $\vec{p} \cdot \vec{b} < 0$ and such that $\vec{p}^T A \geq 0$. As such, we see that if **a)** is not true, then **b)** must be. Thus exactly one of **a)** and **b)** can be true.

Corollary

Let $A_1, \ldots, A_n$ and $\vec{b}$ be a collection of given vectors. Suppose if $\exists \vec{p}$ such that $\vec{p}^T A_i \geq 0$ for all i, then $\vec{p}$ must also satisfy $\vec{p}^T \vec{b} \geq 0$. Then $\vec{b}$ can be expressed as a non-negative linear combination (i.e. a conical combination) of $A_1, A_2, \ldots A_n$.

Proof

We see $\nexists \vec{p}$ such that $\vec{p}^T A \geq 0$ and $\vec{p} \cdot \vec{b} < 0$. As such, we see **a)** from Farkas' Lemma must hold. This implies $\exists \vec{x} \geq 0$ such that $A\vec{x} = \vec{b}$. By matrix multiplication, we see

$$A\vec{x} = \sum_i x_i A_i = \vec{b}$$

so each x_i will be the corresponding coefficient in the conical combination.

Theorem

Suppose $A\vec{x} \leq \vec{b}$ has at least one solution and let d be some scalar. Then the following are equivalent:
(a) Every feasible solution to $A\vec{x} \leq \vec{b}$ satisfies $\vec{c} \cdot \vec{x} \leq d$
(b) $\exists \vec{p} \geq 0$ such that $\vec{p}^T A = \vec{c}^T$ and $\vec{p}^T \vec{b} \leq d$

Proof

We will prove this using a similar LP duality idea to the one we used for Farkas' lemma. Consider the LP below:

$$\text{maximize } \vec{c} \cdot \vec{x}$$

such that

$$A\vec{x} \leq \vec{b}$$

$$\vec{x} \in \mathbb{R}^n$$

Then it has a corresponding dual LP given below:

$$\text{minimize } \vec{p} \cdot \vec{b}$$

such that

$$\vec{p}^T A = \vec{c}^T$$

$$\vec{p} \geq 0$$

If **(a)** holds, then the primal LP is bounded from above. This implies that an optimal solution exists to the primal. By strong duality, this implies there exists a vector $\vec{p}^*$ that is optimal in the dual and there exists $\vec{x}^*$ which is oprimal in the primal such that

$$\vec{p}^* \cdot \vec{b} = \vec{c} \cdot \vec{x}^* \leq d$$

This implies that there exists a feasible $\vec{p}$ (i.e. $\vec{p}^T A = \vec{c}^T$ and $\vec{p} \geq 0$) such that $\vec{p}^* \cdot \vec{b} \leq d$. This means **(a)** implies **(b)**. Now, suppose that **(b)** holds. Then with the weak duality, we see for all primal feasible $\vec{x}$ we have

$$\vec{c} \cdot \vec{x} \leq \vec{p} \cdot \vec{b} \leq d$$

Thus **(b)** implies **(a)**.

Some Topology

Now, we will prove Farkas' Lemma using a second method. We will use a few ideas from topology. If you haven't seen topology before, there are a few important ideas to keep in mind.

1. In a topological space (such as $\mathbb{R}^n$) any closed set will have an open set as a complement (and any open set will have a closed set as a complement)
2. The finite intersection of open sets is open. The union (not necessarily finite) of open sets is open.
3. The intersection (not necessarily finite) of closed sets is closed. The finite union of closed sets is closed

So... what exactly is a closed set in $\mathbb{R}^n$? We will use the following definition of a closed set in $\mathbb{R}^n$. This idea can be extended to other topological spaces (which is outside the scope of this course). A set $S \subseteq \mathbb{R}^n$ if all convergent sequences contained entirely in S must converge to a point in S. Symbolically, if

$$\lim_{i\to\infty} \vec{x}_i \to \vec{x}$$

with $\vec{x}_i \in S$, we must have $\vec{x} \in S$. In particular, this tells us S contains its boundary.

Theorem

All polyhedra are closed in $\mathbb{R}^n$.

Proof

Consider $P = \left\{\vec{x} \in \mathbb{R}^n | A\vec{x} \geq \vec{b}\right\}$. Then for all $\vec{x}_i \in P$, we see

$$A\vec{x}_i \geq \vec{b}$$

If we take the limit of both sides, we see

$$A\vec{x} = A \lim_{i\to\infty} \vec{x}_i = \lim_{i\to\infty} A\vec{x}_i \geq \lim_{i\to\infty} \vec{b} = \vec{b}$$

so $\vec{x} \in P$.

Theorem (Weierstrass' Theorem)

If $f : \mathbb{R}^n \to \mathbb{R}$ is continuous and if S is a non-empty, closed, and bounded subset of $\mathbb{R}^n$, then $\exists \vec{x}^* \in S$ such that

$$f(\vec{x}^*) \leq f(\vec{x}) \ \ \forall \vec{x} \in S$$

Similarly, $\exists \vec{y}^* \in S$ such that

$$f(\vec{y}^*) \geq f(\vec{x}) \ \ \forall \vec{x} \in S$$

We will not prove this. I claim this is a generalization of extreme value theorem. In particular, this theorem doesn't apply for non-closed sets, unbounded sets, if we replace either of these inequalities with strict inequalities, or if f isn't continuous. A good exercise is to find a counterexample when one of these assumptions is removed (but the others are not removed).

Theorem (Separating Hyperplane Theorem)

Let S be a non-empty closed convex subset of $\mathbb{R}^n$ and let $\vec{x}^* \in \mathbb{R}^n$ be a vector not in S. Then $\exists \vec{c} \in \mathbb{R}^n$ such that $\vec{c} \cdot \vec{x}^* < \vec{c} \cdot \vec{x}$ for all $\vec{x} \in S$.

Proof

Let $\|\vec{x}\| = \sqrt{\vec{x} \cdot \vec{x}}$ be the standard Euclidean norm. Fix $\vec{w} \in S$ and let

$$B = \{\vec{x} | \, \|\vec{x} - \vec{x}^*\| \leq \|\vec{w} - \vec{x}^*\|\}$$

and let $D = S \cap B$. (I encourage you to draw a sketch of the relevant vectors and points in$\mathbb{R}^2$ to keep in mind throughout the proof. This isn't perfect, but it will help with intuition). We see $D \neq \emptyset$ as $\vec{w} \in S$ and $\|\vec{w} - \vec{x}^*\| \leq \|\vec{w} - \vec{x}^*\|$ so $\vec{w} \in B$. Further, as B and S are both closed, then $D = S \cap B$ is also closed. Since B is bounded, then it follows that D is also bounded. Finally, consider

$$f(\vec{x}) = \|\vec{x} - \vec{x}^*\|$$

for $\vec{x} \in D$. Then $f(\vec{x})$ is continuous. Since D is non-empty, closed, bounded, and f is continuous, then Weierstrass' theorem applies. There exists $\vec{y} \in D$ such that

$$\|\vec{y} - \vec{x}^*\| \leq \|\vec{x} - \vec{x}^*\| \;\; \forall \vec{x} \in D$$

So $\forall \vec{z} \in S$ such that $\vec{z} \notin D$, we see

$$\|\vec{z} - \vec{x}^*\| > \|\vec{w} - \vec{x}^*\| \geq \|\vec{y} - \vec{x}^*\|$$

So

$$\vec{y} = \text{argmin}_{\vec{x} \in S} f(\vec{x})$$

Now, suppose $\vec{x} \in S$. For all λ such that $0 < \lambda \leq 1$, we see

$$\lambda \vec{x} + (1 - \lambda)\vec{y} = \vec{y} + \lambda(\vec{x} - \vec{y}) \in S$$

as S is convex. This implies

$$\|\vec{y} - \vec{x}^*\|^2 \leq \|\vec{y} + \lambda(\vec{x} - \vec{y}) - \vec{x}^*\|^2$$

$$= \|\vec{y} - \vec{x}^*\|^2 + 2\lambda(\vec{y} - \vec{x}^*) \cdot (\vec{x} - \vec{y}) + \lambda^2 \|\vec{x} - \vec{y}\|^2$$

so

$$0 \leq 2\lambda(\vec{y} - \vec{x}^*) \cdot (\vec{x} - \vec{y}) + \lambda^2 \|\vec{x} - \vec{y}\|^2$$

$$0 \leq 2(\vec{y} - \vec{x}^*) \cdot (\vec{x} - \vec{y}) + \lambda \|\vec{x} - \vec{y}\|^2$$

If we take the limit as $\lambda \to 0$, we see

$$0 = \lim_{\lambda \to 0} 0 \leq \lim_{\lambda \to 0} \left(2(\vec{y} - \vec{x}^*) \cdot (\vec{x} - \vec{y}) + \lambda \|\vec{x} - \vec{y}\|^2\right) = 2(\vec{y} - \vec{x}^*) \cdot (\vec{x} - \vec{y})$$

So

$$0 \leq (\vec{y} - \vec{x}^*) \cdot (\vec{x} - \vec{y}) = (\vec{y} - \vec{x}^*) \cdot \vec{x} - (\vec{y} - \vec{x}^*) \cdot \vec{y}$$

Thus

$$(\vec{y} - \vec{x}^*) \cdot \vec{x} \geq (\vec{y} - \vec{x}^*) \cdot \vec{y} = (\vec{y} - \vec{x}^*) \cdot \vec{x}^* + (\vec{y} - \vec{x}^*) \cdot (\vec{y} - \vec{x}^*)$$

but as $(\vec{y} - \vec{x}^*) \cdot (\vec{y} - \vec{x}^*) = \|(\vec{y} - \vec{x}^*)\|^2 > 0$, we see

$$(\vec{y} - \vec{x}^*) \cdot \vec{x} \geq (\vec{y} - \vec{x}^*) \cdot \vec{x}^* + (\vec{y} - \vec{x}^*) \cdot (\vec{y} - \vec{x}^*) > (\vec{y} - \vec{x}^*) \cdot \vec{x}^*$$

If we set $\vec{c} = \vec{y} - \vec{x}^*$, then we get our result.

Proof 2 of Farkas' Lemma

We will assume the first half of the proof is the same as the first half of the first proof for Farkas' lemma. We will show that if **(a)** from Farkas' lemma doesn't hold, then **(b)** from Farkas' lemma must hold. Suppose $A\vec{x} = \vec{b}$ has no solution for any $\vec{x} \geq 0$. Consider the set,

$$S = \{A\vec{x} | \vec{x} \geq 0\} = \{\vec{y} | \exists \vec{x} \text{ such that } \vec{y} = A\vec{x}, \vec{x} \geq 0\}$$

Suppose $\vec{b} \notin S$. We see S is a convex cone. Further, we see S is closed as it is the projection of the polyhedron:

$$P = \{(\vec{x}, \vec{y}) \, | \vec{y} = A\vec{x}, \vec{x} \geq 0\}$$

onto it's $\vec{y}$ coordinate. Now, as $\vec{b} \notin S$, separating hyperplane theorem tells us $\exists \vec{p}$ such that $\vec{p} \cdot \vec{b} < \vec{p} \cdot \vec{y}$ for all $\vec{y} \in S$. Since $\vec{0} \in S$, this implies $\exists \vec{p}$ such that

$$\vec{p} \cdot \vec{b} < \vec{p} \cdot \vec{0} = 0$$

Further, for all A_i and $\lambda > 0$, we have $\lambda A_i \in S$ (as S is a cone). We also have

$$\vec{p} \cdot \vec{b} < \lambda \vec{p} \cdot A_i \ \forall i$$

If we divide by λ, we see

$$\frac{\vec{p} \cdot \vec{b}}{\lambda} < \vec{p} \cdot A_i \ \forall i$$

If we let $\lambda \to \infty$, we see

$$0 = \lim_{\lambda \to \infty} \frac{\vec{p} \cdot \vec{b}}{\lambda} \leq \lim_{\lambda \to \infty} \vec{p} \cdot A_i = \vec{p} \cdot A_i \ \forall i$$

Thus

$$\vec{p}^T A \geq 0$$

and we have shown Farkas' Lemma.